

System Analysis of Itqan

IS498 | Track A | Independent draft | 22 September 2026

1. Purpose and analysis basis

Itqan is a proposed standalone Quran recitation coach for Juz Amma. It supports independent practice through voice analysis and feedback, while allowing teachers to follow enrolled learners' stored learning results. The initial scope is 37 Surahs, from An-Naba to An-Nas. The application supports learning; it does not establish that a recitation is religiously correct or replace a qualified teacher. [2]

This analysis was developed directly from the project requirements, current data model, explicit identity/enrollment decisions, transient-audio policy, and test plan. No earlier system-analysis document or derivative draft was used. The newly drawn models describe intended behavior and are not evidence of implementation. [2]-[6]

The handbook requires the analysis to connect actors, scenarios, process logic, data, and object interactions. This chapter follows Section 13.2 and explains each model. Appendix R inconsistently labels some explicitly required diagrams as recommended; the activity, sequence, and class models are retained here. [1, pp. 18-19, 75-77]

1.1 Problem and system boundary

The project starts from the hypothesis that learners practising alone may not notice pronunciation or Tajweed errors and may lack timely guidance. User research is still outstanding, so these statements are a rationale to investigate, not reported research findings. No claim about competitor limitations is needed to define the system. [2, 7]

Table 1: Scope of the proposed system

Included	Excluded or conditional
Accounts; Surah selection; learner modes; active recitation feedback; results and progress.	Full-Quran coverage, integration with another Quran app, and public third-party APIs are future work.
Teacher dashboard; class management and code-based enrollment.	No Admin actor or guardian account is introduced.
Corrective reference audio; encouragement in Al-Mushajji.	Smart Prompting, assignments, rewards, and challenges require explicit scope decisions.

Table 1 separates the current requirements from extensions. The body is a standalone analysis chapter; detailed scenarios and test mappings are placed in the appendices. It is not a complete graduation-project report.

2. Stakeholders and system services

Learners and teachers are the primary human actors. The identity provider is a supporting external actor. The AI analysis capability is inside Itqan's logical boundary, even when deployed separately. The project team and academic supervisors are stakeholders rather than application users. Table 2 relates each stakeholder to an observable need.

Table 2: Stakeholder needs and system responsibilities

Stakeholder	Need	Responsibility
Learner	Practise and interpret correction.	Show verses, capture consented audio, locate supported errors, present feedback and progress.
Teacher	Follow learners without unnecessary access.	Restrict dashboard and messaging to owned classes and active enrollments.
Project team	Build and evaluate a feasible product.	Define model coverage, data rules, failure handling, and repeatable acceptance tests.
Academic supervisors	Assess a defensible analysis and design.	Keep requirements, models, evidence, and limitations traceable.

2.1 Meaning of the three modes

Al-Mujawwid configures correction-oriented learner practice (FR-08). **Al-Mushajji** adds encouraging interaction to practice (FR-09); rewards and challenges are not assumed. **Al-Mu'allim** groups teacher capabilities (FR-10), realized through dashboard, classes, and optional messaging. It is not treated as an AI actor or a learner recitation-session goal.

2.2 Information entering and leaving the system

Learners supply account details, a Surah and mode choice, session commands, recitation audio, and class join codes. Itqan returns Quran content, feedback, eligible results, progress, and messages. Teachers supply class details, monitoring selections, and optional messages; they receive authorized learning summaries. The identity provider exchanges authentication outcomes with Itqan. Trusted Quran text and reference audio are input content assets, not generated corrections.

Recitation audio exists only during processing and is discarded after analysis, including failure paths. Stored results may describe an error and its location but must not contain raw learner audio or a recording link. Corrective reference audio is a different asset. [3]

3. Use-case model

Figure 1 groups learner goals into six use cases. A use case represents an actor's goal rather than an internal processing step. UC-L3 therefore contains capture, analysis, error classification, localization, feedback, and finalization. Written scenarios for every displayed use case appear in Appendix A.

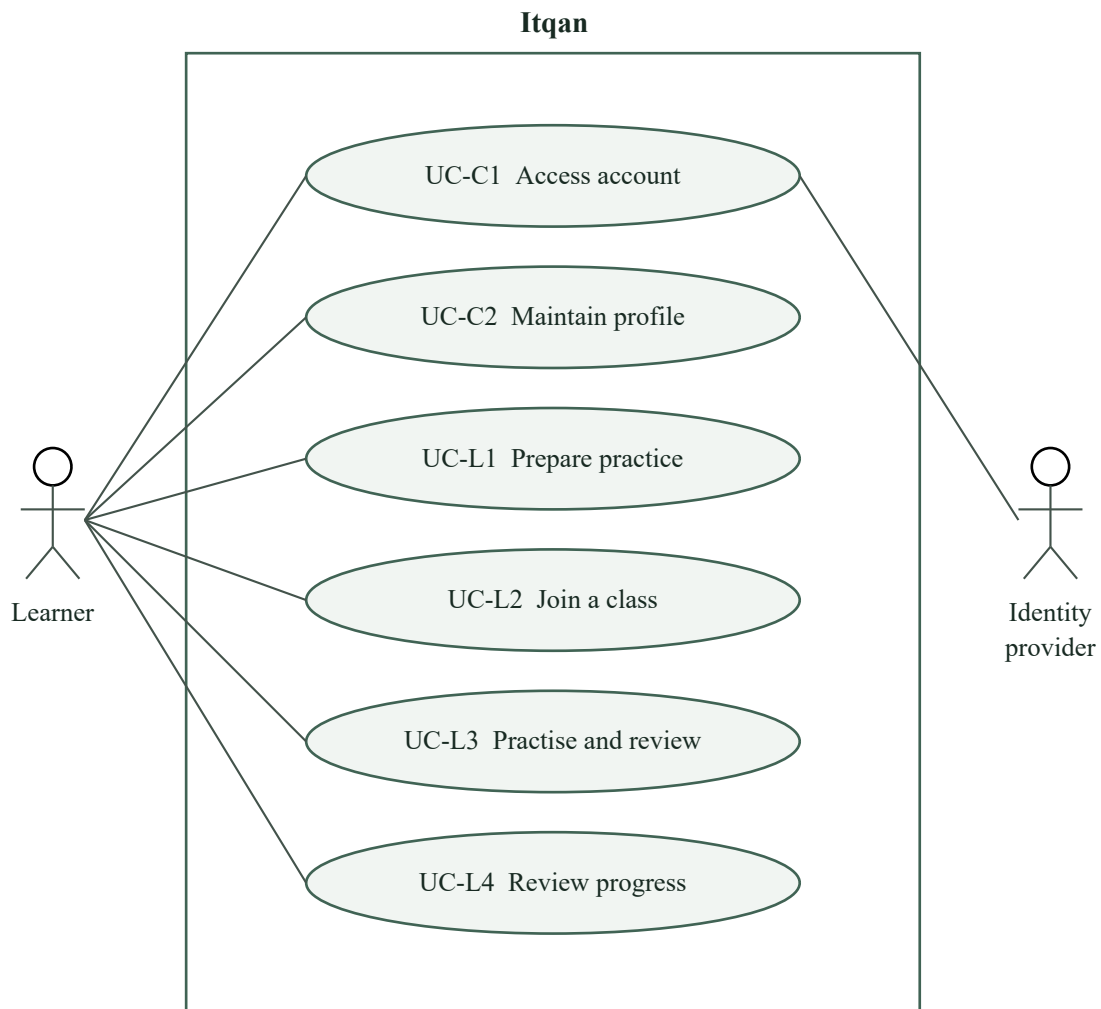


Figure 1: Learner use cases; plain lines indicate actor associations

The provider participates only in account access. Successful authentication is a precondition for protected use cases, rather than a repeated included use case. Corrective audio is an optional step of UC-L3. Al-Mujawwid and Al-Mushajji are session configurations, not separate actors. UC-L2 follows the documented join-code decision rather than assuming that a teacher silently enrolls a learner. [4]

3.1 Teacher use cases and business rules

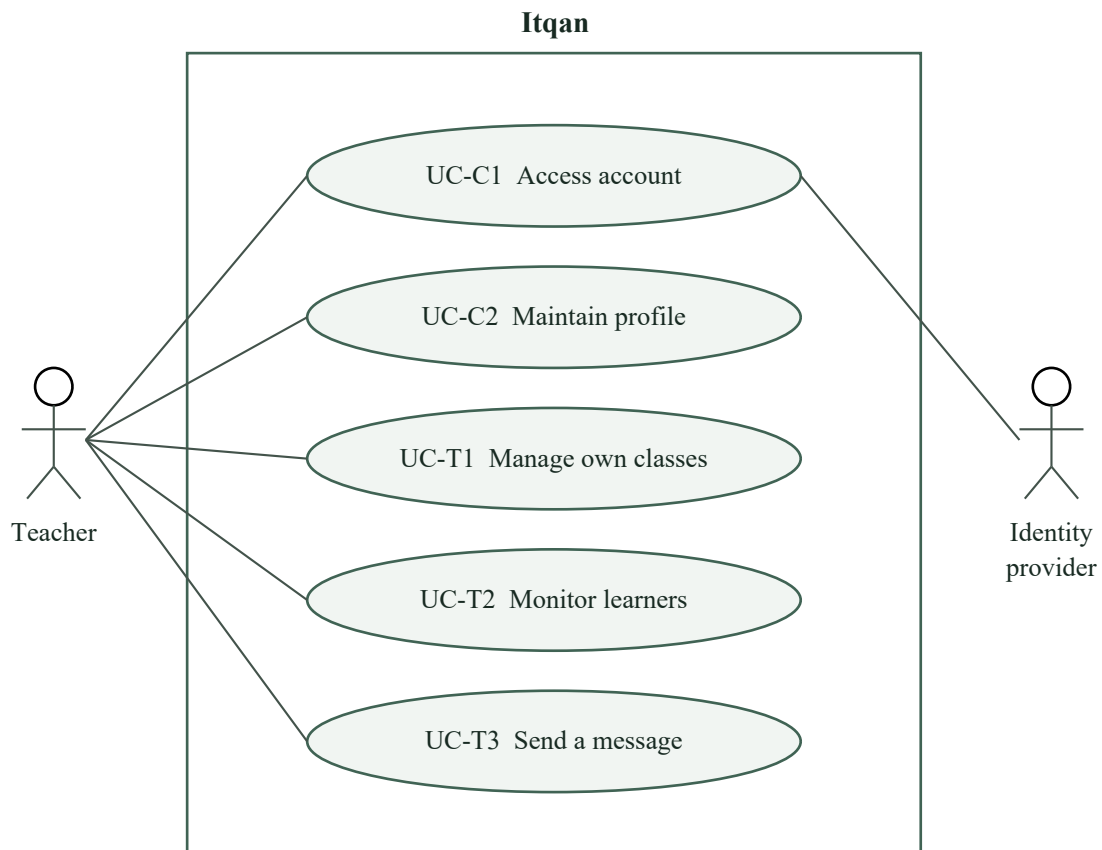


Figure 2: Teacher goals; UC-T3 is a Could Have requirement

Figure 2 separates monitoring from class management. A teacher can have a dashboard requirement even if more extensive class-management actions are deferred. FR-23 is Must Have although its umbrella FR-10 is Should Have; these existing priorities must be reconciled explicitly. [2]

Table 3: Rules that constrain every applicable scenario

Rule	Effect
BR-1 Scope and ordering	Select a supported Surah before learner mode and session start.
BR-2 Access	Teacher role, class ownership, and active enrollment are all necessary.
BR-3 Audio lifecycle	Do not persist learner audio in files, logs, databases, backups, or messages.
BR-4 Identity and consent	Store provider identity and per-purpose consent history; no local password hash.
BR-5 Results	Do not invent a mastery formula or report failed analysis as successful mastery.

Table 3 combines explicit decisions (BR-1 to BR-4) with an analysis safeguard (BR-5) that must be reflected in the scoring specification.

4. Process model: learner practice

Figure 3 shows the proposed control flow for UC-L3. Decisions are explicit so that unavailable analysis and a learner's request to continue do not lead accidentally to successful completion.

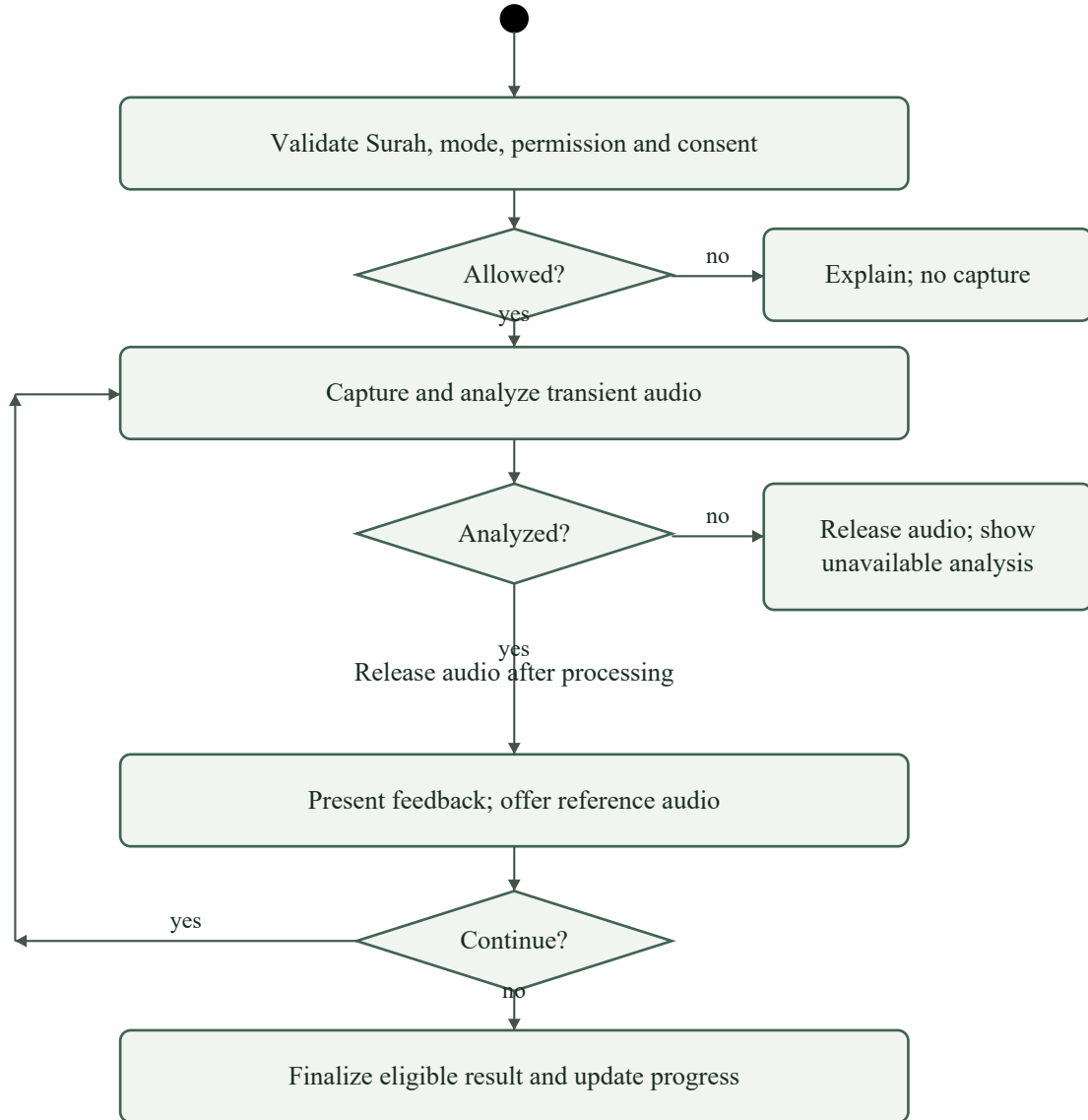


Figure 3: Learner activity flow; retry/recovery details remain proposed

FR-11 to FR-18 govern the active loop. Audio buffers are released once their analysis finishes or fails. The failure branch never enters the valid-result path automatically; the learner may end or repeat practice under an approved recovery policy. Finishing stops capture before finalization. Only qualifying completed results update progress under the current class-model rule. [3, 5]

Per-verse processing is a design option, not proof of real-time compliance. Segmentation must allow recitation without a forced stop after each word or verse, as NFR-03 requires. The result formula, pending-analysis policy, and completion eligibility still need agreed definitions.

4.1 Process model: teacher monitoring

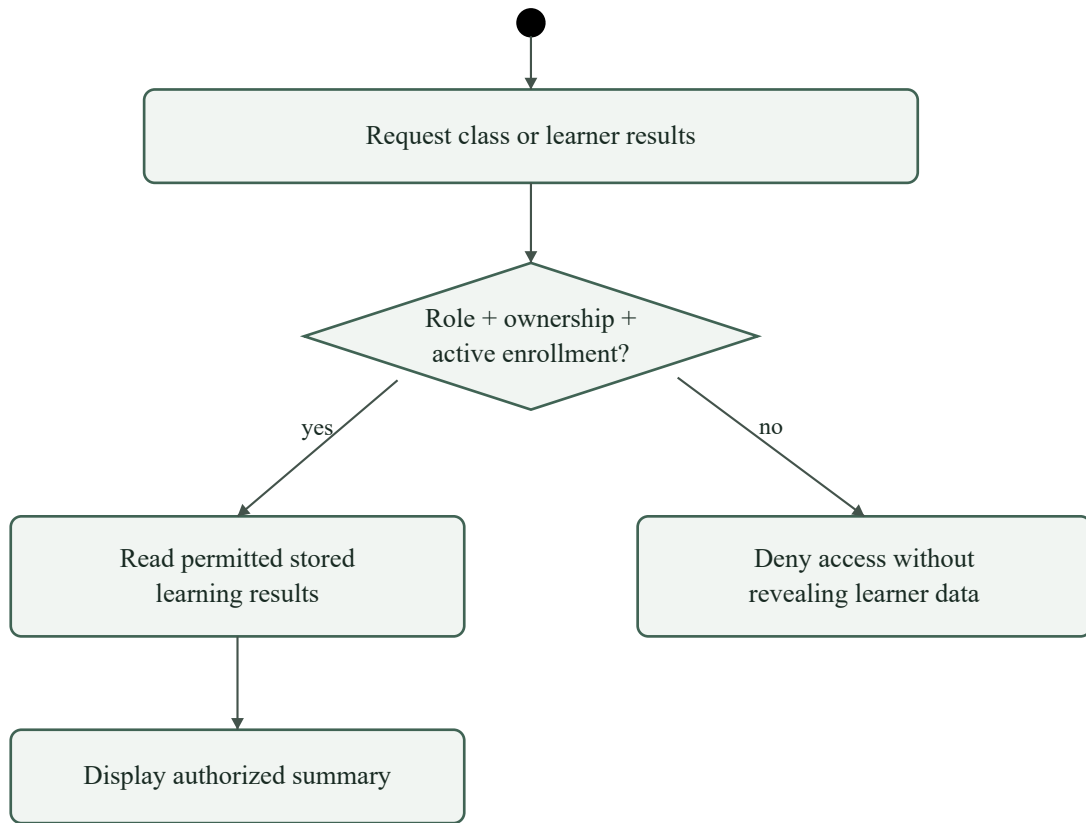


Figure 4: Teacher monitoring activity with an explicit denied-access branch

Figure 4 makes authorization a server/data-access responsibility. Hiding a learner in the interface is insufficient: every monitoring request must recheck the teacher's permitted scope. Revoked or unrelated enrollment reaches the deny branch without retrieving the protected result. [5, 6]

4.2 Alternative flows and session lifecycle

Table 4: Failure handling required by the analysis

Situation	Required outcome or proposed response
Microphone permission denied	Do not capture; explain how to return after permission is granted.
Analysis failure	Discard audio; show an unavailable result. Retry behavior remains to be specified.
Interrupted finalization	Preserve permitted metadata; do not claim progress was saved without confirmation.
Unknown or inactive class code	Do not create active enrollment; return a usable correction path.
Enrollment ends	Deny subsequent teacher reads and message sends.
No practice history	Display an empty state, not an inferred mastery score.

Table 4 extends the normal paths with outcomes linked to ET-01 to ET-05 and privacy tests. These are requirements for future implementation, not results of executed tests.

Session lifecycle

The current schema distinguishes active, completed, abandoned and failed sessions. Starting creates an active session. An eligible end operation completes it; unrecoverable processing failure marks failure; an unfinished attempt may be abandoned under a policy still to be defined. Capture or further analysis against a terminal session must be rejected. [5]

A completed session has an end time, whereas an active session does not. The distinction between stopping early and completing the intended scope needs a documented completion rule. A successful save should publish the session outcome and corresponding progress consistently; a repeated finalization request must not count a second practice attempt.

These transitions explain the integrity expected from RecitationService and RecitationSession. They are a textual lifecycle specification, not an assertion that the recovery policy or every transition is approved. The acceptance suite must exercise repeated end commands, late analysis responses, and application interruption.

5. Interaction model: learner session

Figure 5 allocates the learner workflow to logical responsibilities. The application handles presentation and capture; the recitation coordinator manages session state; analysis returns derived findings; persistence receives metadata and results only.

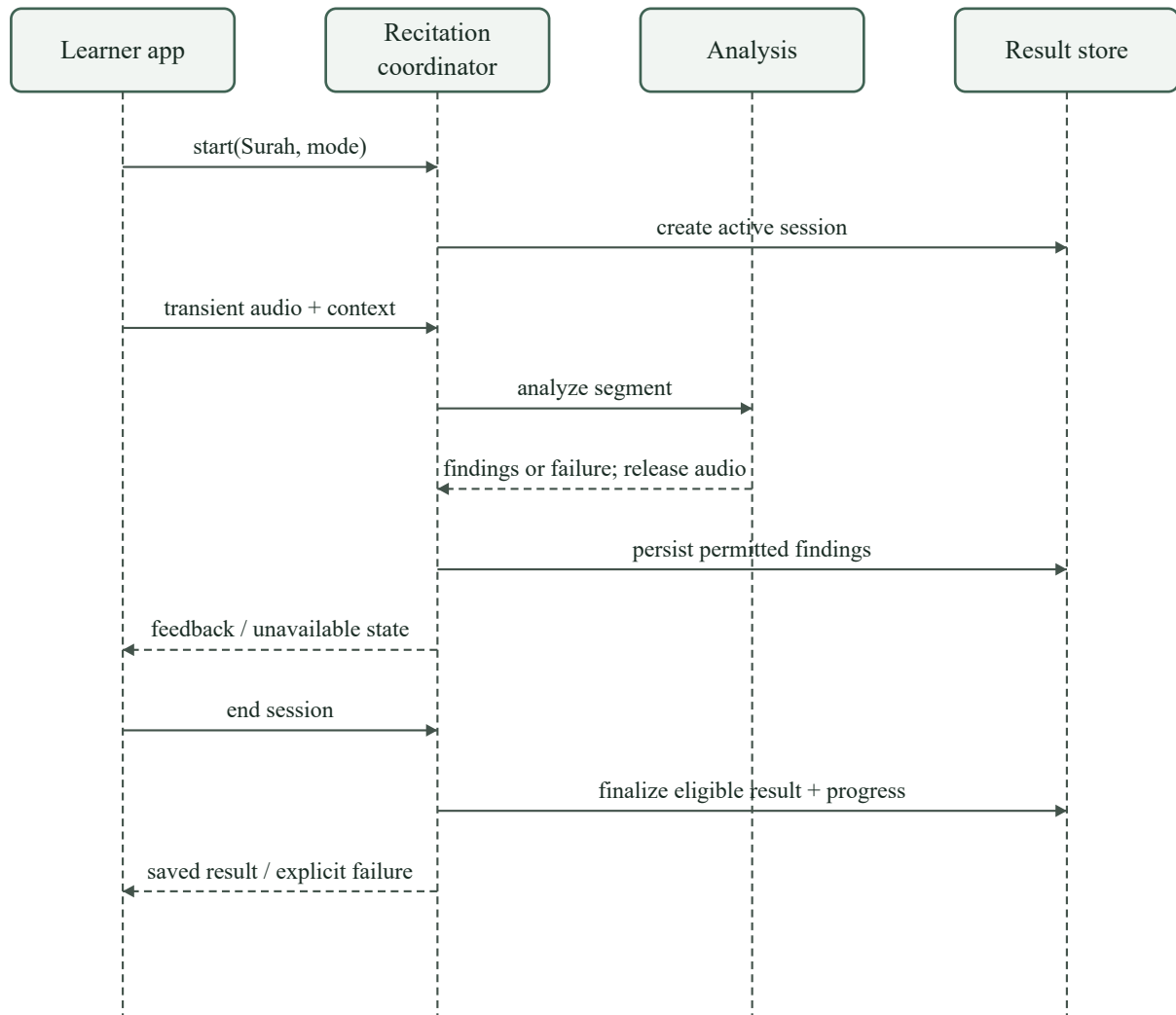


Figure 5: One session interaction; segment messages repeat during active practice

The diagram represents the successful segment path and names the failure return. If analysis fails, the coordinator skips persistence of successful findings and sends an unavailable state. The client, coordinator, and analysis component all release their audio copies when processing ends. Corrective playback uses a trusted reference, not stored learner audio.

Finalization must use the approved score policy. A proposed idempotent completion operation prevents repeated end requests from duplicating mastery history or daily practice. No valid score should be calculated from missing evidence by default. These integrity rules refine NFR-05 and require implementation tests.

5.1 Interaction model: enrollment and monitoring

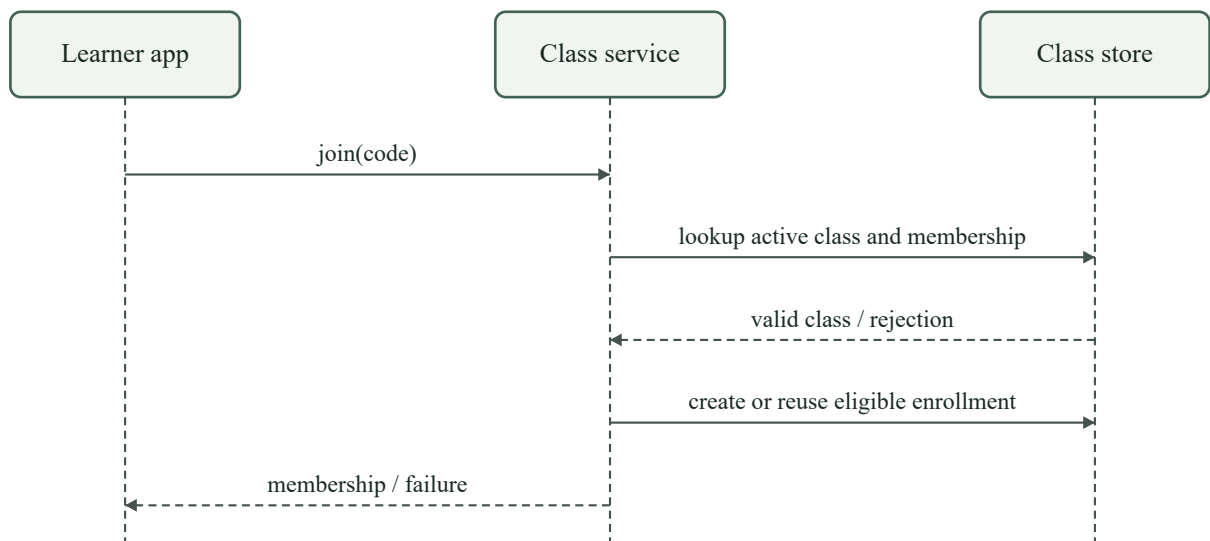


Figure 6: Class enrollment; invalid lookup terminates before membership creation

Figure 6 implements the code-based decision. Existing membership should be handled without duplicate enrollment, consistent with the unique class/learner key. Join-code rotation and rejoining after removal remain unresolved. [4, 5]

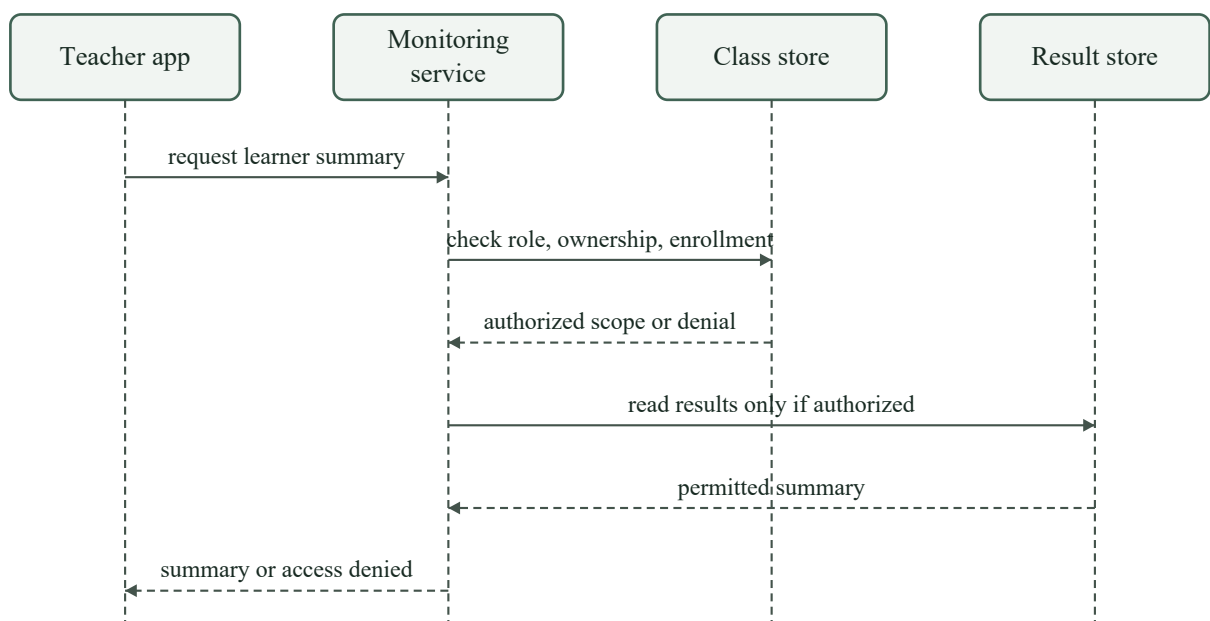


Figure 7: Teacher monitoring; denied authorization bypasses the result store

Figure 7 corresponds to UC-T2 and PT-03/PT-04. UC-T3 repeats the same authorization before saving a message. No teacher-facing response includes learner audio, recording URLs, transcripts, or contact details. [6]

6. Data model: identity and learning context

The data model separates identity, content, practice evidence, and teacher relationships. Figures 8-10 are newly drawn relational projections of the project schema. Numbers at line ends are multiplicities; repeated boxes identify the same relation. They are intentionally split to remain readable on A4. [5]

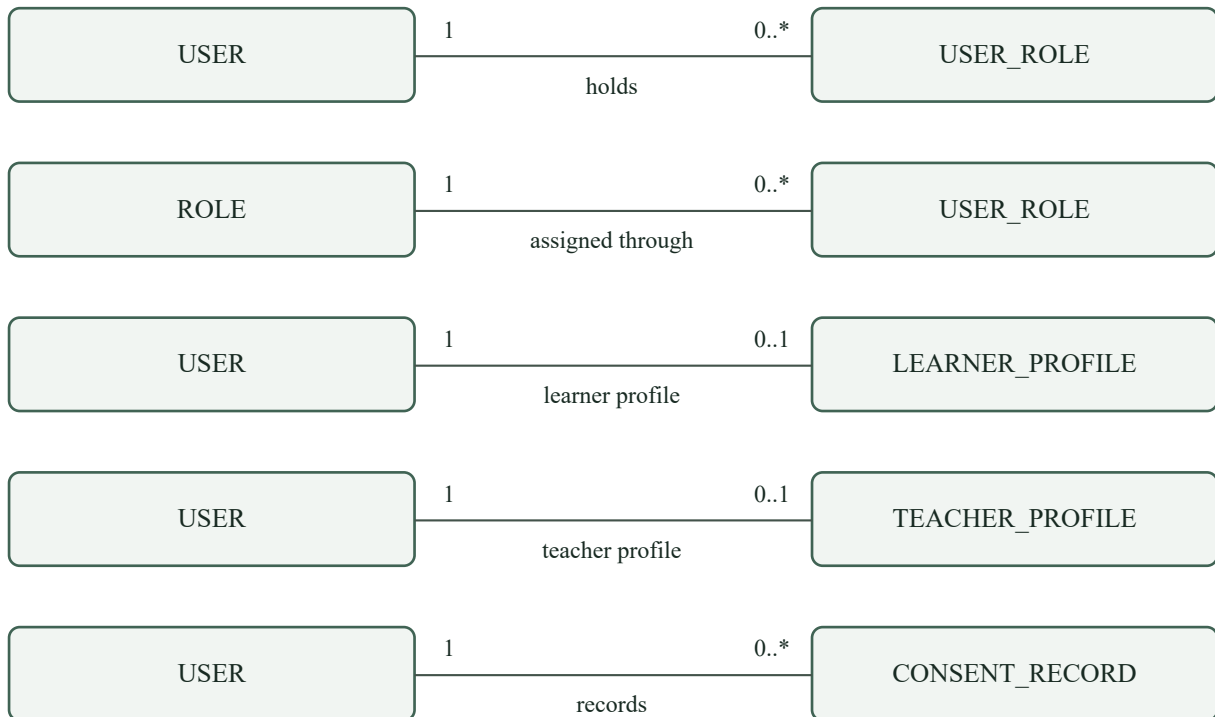


Figure 8: Identity relations; one account may hold both roles

Table 5: Key relational constraints for identity and content

Relation	Key and meaning
USER	PK user_id; unique identity_subject and email. Authentication is delegated.
USER_ROLE	Composite PK (user_id, role_id); both columns are foreign keys.
CONSENT_RECORD	PK consent_id; FK user_id. One active record per user and purpose.
SURAH / VERSE	VERSE references SURAH; (surah_id, verse_number) is unique.
VERSE_WORD	FK verse_id; unique (verse_id, word_position) anchors localized errors.

Table 5 summarizes essential keys. The companion `RELATIONAL_SCHEMA.md` retains the full existing field dictionary and constraints; it is copied from the data-model source, not derived from a prior analysis.

6.1 Data model: recitation evidence

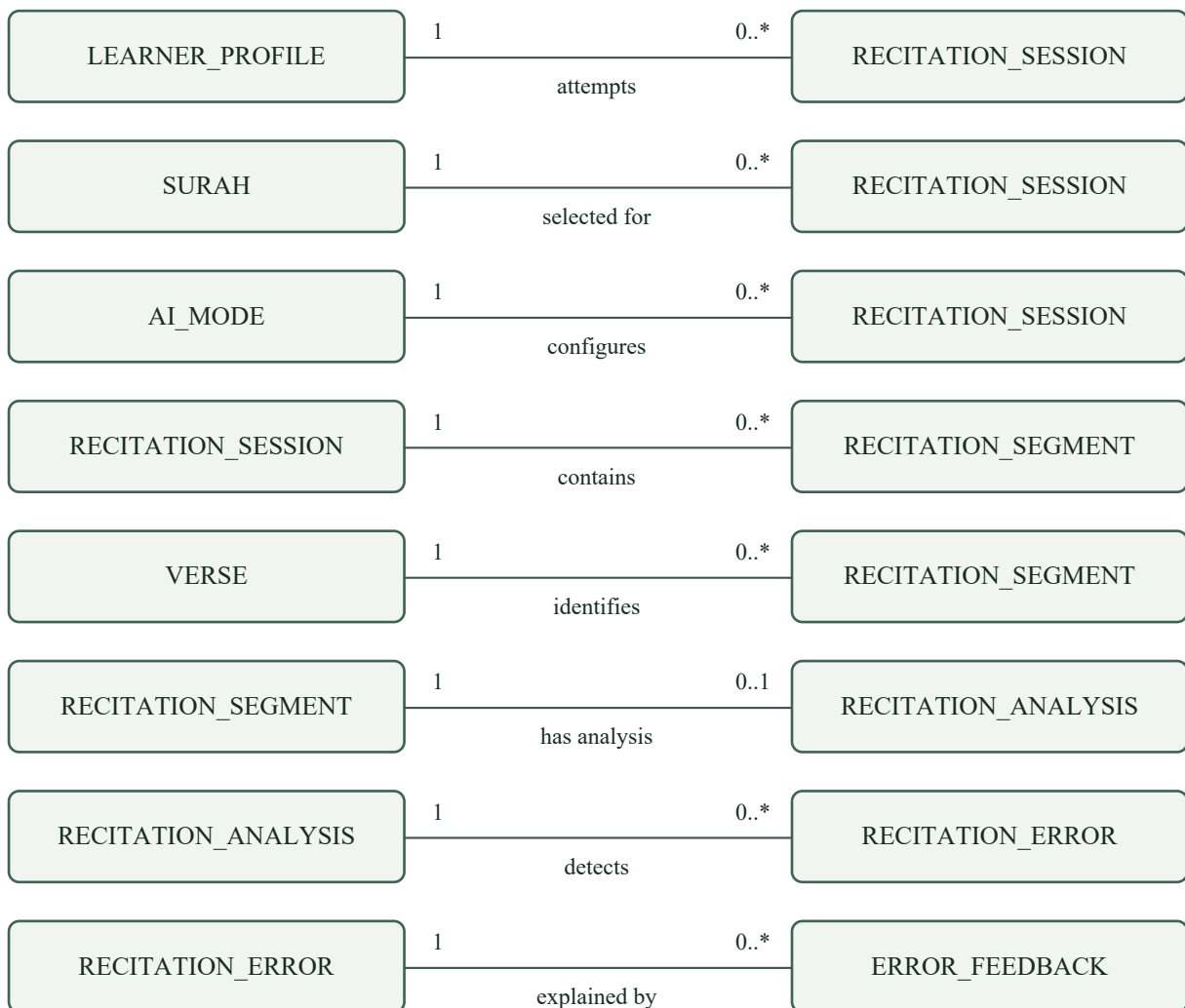


Figure 9: Core recitation evidence and result relationships

Figure 9 stores the evidence of a practice attempt without storing its audio. **ERROR_TYPE** classifies each **RECITATION_ERROR**; **VERSE_WORD** optionally anchors it. A null word reference must not be presented as a precisely localized error. FR-15 requires a defined location policy for ambiguous or unaligned findings.

RECITATION_SEGMENT stores timing offsets and permitted derived fields; these are not recording locations. **RECITATION_ANALYSIS** records model/version and outcome for evaluation traceability. **AUDIO_ASSISTANCE** points to reference content and exactly one error or pause. **PAUSE_EVENT** and pause-triggered assistance must not make unapproved Smart Prompting a current requirement.

The mode catalogue contains all three project modes, but learner-session creation must restrict selection to learner modes. A database row alone does not authorize a workflow. Scores must lie in 0-100 and confidence in 0-1; these bounds do not define the score formula.

6.2 Data model: progress and teacher access

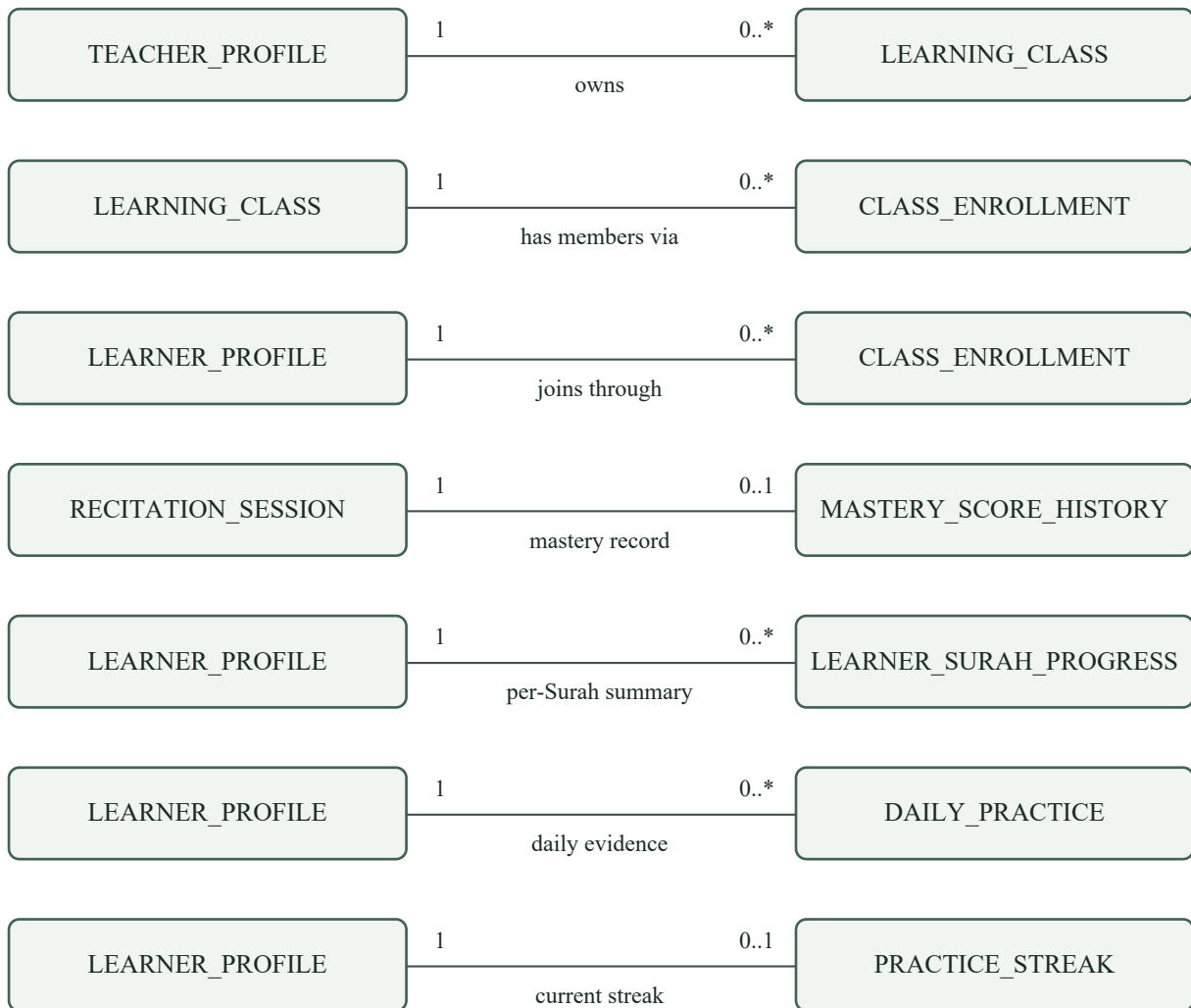


Figure 10: Membership and derived progress relations

Figure 10 distinguishes per-session history from cumulative summaries. **LEARNER_SURAH_PROGRESS** is unique per learner/Surah and **CLASS_ENROLLMENT** per class/learner. Streak calculation uses daily evidence, but the qualifying practice and day-boundary rules are still open. **TEACHER_MESSAGE** references its sender and learner, with optional class/session context. [5]

Proposed assignments, performance snapshots, challenges, and rewards remain outside the core analysis. The companion schema preserves their explicit proposed status. No requirement is inferred merely from the existence of an optional table.

The full schema includes JUZ, Quran reference and rule catalogues, error types, pause evidence, feedback, assistance, and the optional extensions. These are not omitted from the dictionary; the printed projections focus on the relationships needed to explain the core workflows.

7. Class model and responsibility allocation

Figure 11 is an analysis-class projection rather than a database diagram. Entity classes hold state; services coordinate user goals; interfaces isolate capture, analysis, and storage. Attributes and operations are representative, while the complete existing class model remains the detailed design reference. [5]

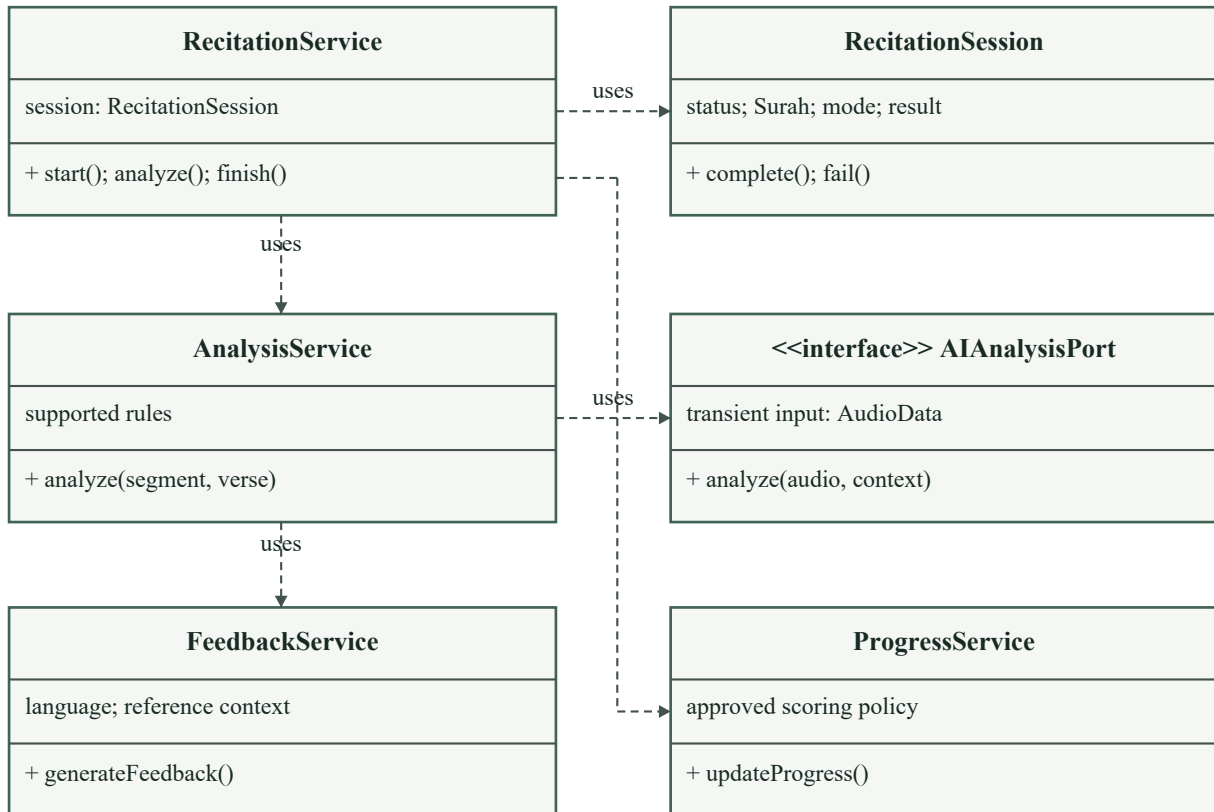


Figure 11: Recitation class responsibilities; dashed arrows are dependencies

RecitationService uses an AudioCapturePort to obtain transient AudioData and a RecitationRepository to persist permitted state. AnalysisService uses AIAnalysisPort; the service boundary does not imply an external commercial provider. FeedbackService translates supported findings into usable feedback. ProgressService updates derived summaries only from eligible completed results.

AuthenticationService coordinates provider identity and account/profile records. ClassService owns class/enrollment operations. TeacherMonitoringService and MessagingService enforce scope before using repositories. Domain entities correspond to persisted relations; services, repositories, ports, and AudioData have no database tables.

USER maps to User, RECITATION_SESSION to RecitationSession, and other relation names map to their corresponding domain classes in the class-model source. One account can own learner and teacher profiles; the roles are not modeled as mutually exclusive subclasses. Optional assignment and gamification classes remain proposed.

8. Quality requirements and unresolved decisions

The non-functional requirements constrain the analysis, but most lack numerical acceptance thresholds. Table 6 defines how quality will be observed without inventing targets. The existing test identifiers are retained. [2, 6]

Table 6: Quality obligations and planned evidence

Requirements	Evidence	Decision still needed
NFR-01 Accuracy	NFT-01: expert-labelled precision, recall and location accuracy.	Supported rules, evaluation sample and pass thresholds.
NFR-02 / NFR-03 Timing	NFT-02/03: end-to-end latency and uninterrupted recitation.	Latency target, segmentation, load and network conditions.
NFR-04 Usability	NFT-04: task completion, errors and assistance.	Participant profile and acceptance threshold.
NFR-05 Reliability	NFT-05 / ET-03: interruption and recovery evidence.	Finalization, retry and incomplete-session rules.
NFR-06 / NFR-07	PT-01 to PT-07: access, consent and no retained learner audio.	Provider details, non-audio retention and withdrawal effects.
NFR-08 / NFR-09	NFT-06/07: extension and modularity review.	Capacity assumptions and approved extension scope.
NFR-10 / NFR-11	NFT-08/09: device compatibility, Arabic/English and RTL checks.	Supported device and platform matrix.

8.1 Decisions that block acceptance

The mastery formula (OQ-08), supported detection approach and benchmark (OQ-09/OQ-10), and real-time behavior (OQ-11) are necessary before score and AI-quality tests can pass. OQ-07 must define the difference between learner modes and whether Smart Prompting is included. Dashboard detail and class-management actions remain open, although join-code enrollment is documented. [4, 7]

Identity-provider selection, deployment, approved Quran/reference-audio assets, and processing arrangements require separate design confirmation. User research must provide actual instruments and findings. This chapter does not assert approvals, user-study results, model accuracy, or a working application.

Implementation readiness requires approved rules, aligned models and testable acceptance criteria. The report can be reviewed now; unresolved choices should remain visible until the team records a decision.

Appendix A. Written use-case scenarios

UC-C1: Access account

Actor: Learner or teacher; managed identity provider

Requirements: FR-01, FR-02

Trigger/preconditions: User chooses registration or sign-in; provider is reachable.

Main success path: 1. Begin provider authentication. 2. Validate its returned identity. 3. Create or locate the permitted application account. 4. Load authorized roles and open the appropriate area.

Alternatives and exceptions: Invalid authentication or unavailable provider gives no protected access. Role assignment and account-recovery policy remain open.

Postcondition: Authenticated application session, or explicit unsuccessful access. No local password is stored.

Verification: FT-01, FT-02, PT-07

UC-C2: Maintain profile

Actor: Authenticated learner or teacher

Requirements: FR-03

Trigger/preconditions: User opens their own profile.

Main success path: 1. Retrieve authorized fields. 2. User changes permitted values. 3. Validate and save. 4. Confirm the saved profile.

Alternatives and exceptions: Reject unauthorized changes; explain invalid fields. A failed save must not be displayed as successful. Editable-field scope requires approval.

Postcondition: Only authorized profile data changes.

Verification: FT-03

Appendix A. Scenarios (continued)

UC-L1: Prepare practice

Actor: Learner

Requirements: FR-04, FR-05, FR-06, FR-07

Trigger/preconditions: Learner opens practice; supported Quran content is available.

Main success path: 1. Browse Juz Amma. 2. Select a Surah. 3. Display its verses. 4. Choose Al-Mujawwid or Al-Mushajji. 5. Proceed to the configured practice screen.

Alternatives and exceptions: An empty search is explained. Missing content prevents that selection; generated Quran text is not substituted. Changing the Surah updates the context.

Postcondition: Surah and mode are ready; microphone capture has not started.

Verification: FT-04, FT-05, FT-06, ET-04

UC-L2: Join a class

Actor: Authenticated learner

Requirements: FR-24 refinement; identity/enrollment decision

Trigger/preconditions: Learner enters a class join code.

Main success path: 1. Validate the code and class state. 2. Check existing membership. 3. Create eligible active enrollment. 4. Confirm membership.

Alternatives and exceptions: Invalid/inactive codes do not enroll. A duplicate join should reuse existing membership. Rejoining a removed class needs an explicit policy.

Postcondition: Enrollment exists only for the intended class and learner.

Verification: FT-14, ET-05; add join-code assertions

Appendix A. Scenarios (continued)

UC-L3: Practise and review

Actor: Authenticated learner

Requirements: FR-08, FR-09, FR-11, FR-12, FR-13, FR-14, FR-15, FR-16, FR-17, FR-18, FR-19, FR-20

Trigger/preconditions: Surah/mode selected; applicable consent, microphone permission and analysis connectivity available.

Main success path: 1. Start session and capture. 2. Analyze against expected content. 3. Classify supported errors and locate affected words. 4. Discard audio after analysis. 5. Present feedback and optional reference playback. 6. Continue as requested. 7. Stop capture when ending. 8. Finalize eligible results with the approved formula, save, and display mastery plus detected words.

Alternatives and exceptions: Permission denial prevents capture. Failure or uncertain analysis must not become a fabricated correction or perfect score. Explain unavailable analysis and offer the approved recovery path. Al-Mushajji adds encouragement; detailed behavior remains open.

Postcondition: Completed result and permitted metadata, or an explicit failed/incomplete outcome; no retained audio.

Verification: FT-07 to FT-11, PT-01/02/06, ET-01/02/03

UC-L4: Review progress

Actor: Authenticated learner

Requirements: FR-21, FR-22

Trigger/preconditions: Learner opens progress.

Main success path: 1. Retrieve own eligible history. 2. Present fully mastered Surah count and average mastery. 3. If included, calculate/show streak from qualifying practice days.

Alternatives and exceptions: An empty history is not a zero-ability assessment. Average-mastery aggregation, full mastery and streak day-boundary rules require definition.

Postcondition: Learner sees available cumulative results derived from permitted history.

Verification: FT-12

Appendix A. Scenarios (continued)

UC-T1: Manage own classes

Actor: Authenticated teacher

Requirements: FR-10, FR-24

Trigger/preconditions: Teacher creates or changes an owned class.

Main success path: 1. Enter allowed class details. 2. Validate the teacher scope and input. 3. Save the class. 4. Provide its unique join code for learner-initiated enrollment.

Alternatives and exceptions: Deny changes to another teacher's class. Do not claim a failed save succeeded. Archival, removal and code rotation need agreed behavior.

Postcondition: Authorized class changes persist; no learner is silently enrolled.

Verification: FT-14, PT-07

UC-T2: Monitor learners

Actor: Authenticated teacher

Requirements: FR-10, FR-23

Trigger/preconditions: Teacher opens dashboard or selects a learner.

Main success path: 1. Verify role, class ownership and active enrollment. 2. Determine permitted result scope. 3. Read stored activity, completion, mastery and progress. 4. Display the authorized summary. Recheck access for subsequent requests.

Alternatives and exceptions: Unrelated, ended or revoked enrollment denies access before protected retrieval. An authorized learner without results has an empty state.

Postcondition: Only authorized learning results are disclosed; no audio, transcript or contact details.

Verification: FT-13, PT-03/04/05, ET-05

UC-T3: Send a message

Actor: Teacher; enrolled learner receives

Requirements: FR-10, FR-25 (Could Have)

Trigger/preconditions: Optional messaging is included; teacher selects an eligible learner.

Main success path: 1. Enter message. 2. Validate content and current enrollment scope. 3. Save authorized message. 4. Make it available to the intended learner.

Alternatives and exceptions: Revoked scope or save failure produces no success confirmation. Reply threads, push notifications and read-receipt behavior are not implied.

Postcondition: Permitted message is stored or failure is explicit.

Verification: FT-15; add send-time authorization test

Appendix B. Functional traceability

Table 7 covers every functional requirement in the working baseline. Requirements are grouped where they support the same actor goal. Priorities are inherited rather than reapproved: 17 Must, 6 Should, and 2 Could. The primary stakeholders are learners for learner rows, teachers for teacher rows, and both roles for account rows. [2]

Table 7: Requirements to scenarios, responsibilities and tests

Requirements	Scenario / responsibility	Test evidence
FR-01, FR-02	UC-C1 / AuthenticationService	FT-01, FT-02
FR-03	UC-C2 / account/profile handling	FT-03
FR-04, FR-05, FR-06, FR-07	UC-L1 / QuranCatalogService, session setup	FT-04 to FT-06
FR-08, FR-09	UC-L3 / AnalysisService, mode-sensitive feedback	FT-07, FT-08
FR-10	UC-T1 to UC-T3 / teacher capabilities umbrella	FT-13 to FT-15
FR-11, FR-12	UC-L3 / RecitationService, AudioCapturePort	FT-09, PT-01/02/06
FR-13, FR-14, FR-15, FR-16	UC-L3 / analysis, location, FeedbackService	FT-07, ET-02
FR-17, FR-18	UC-L3 / reference playback, continuation	FT-10
FR-19, FR-20	UC-L3 / finalization and result presentation	FT-11, ET-03
FR-21, FR-22	UC-L4 / ProgressService	FT-12
FR-23	UC-T2 / TeacherMonitoringService	FT-13, PT-03/04/05
FR-24	UC-T1, UC-L2 / ClassService	FT-14, ET-05
FR-25	UC-T3 / MessagingService	FT-15

Traceability is not test execution. Tables 6 and 7 identify planned verification, while scoring, mode, class-management and real-time decisions still block some assertions. The join-code flow requires updates to older FR-24 wording and FT-14 steps.

Appendix C. Consistency and source register

Table 8: Cross-model review of the new analysis

Check	Result and remaining work
Actors and scenarios	Every goal in Figures 1-2 has a written scenario. Supporting identity provider is identified.
Main workflows	Learner and teacher activities are explained; learner, enrollment and monitoring interactions have sequences.
Data and classes	Printed views are explicit projections of the shared schema/class sources. Full companion schema preserves all relations and proposed labels.
Requirements and tests	All FR-01 to FR-25 and NFR-01 to NFR-11 are mapped; no passing implementation tests are claimed.
Prototype	Account, practice, progress and class views must serve the mapped goals. Messaging/entry and failure-state coverage needs prototype verification.
Submission	A4 portrait, 2.5 cm margins and readable vector diagrams are used. Add actual team number and integrate with the complete report's numbering, front matter and 30-page body budget.

Table 8 records the limits of this analysis rather than awarding compliance by assumption. The handbook requires the full report to contain research evidence, architecture justification, a prototype and its evaluation, test planning, and other project sections. This chapter covers the system-analysis contribution. [1]

References

[1] King Saud University, Department of Information Systems, Graduation Project Handbook, user-supplied 77-page PDF, reviewed 22 September 2026. Section 13.2, pp. 18-19; Section 14.2, p. 21; Sections 24-27, pp. 32-33; Appendix R, pp. 75-77. Publication date not established.

[2] Itqan project requirements, docs/project/REQUIREMENTS.md. Working requirements and priorities, reviewed 22 September 2026.

[3] Transient Audio Policy Alignment, docs/superpowers/specs/2026-09-19-transient-audio-policy-design.md.

[4] Artifact Consistency Design, docs/superpowers/specs/2026-09-20-artifact-consistency-design.md.

[5] Itqan relational schema and class-model notes, docs/database/RELATIONAL_SCHEMA.md and docs/diagrams/class/CLASS_DIAGRAM_NOTES.md; reviewed 22 September 2026.

[6] Requirement-Linked Test Plan, docs/ui-ux/REQUIREMENT_LINKED_TEST_PLAN.md; planned tests, not execution results.

[7] Itqan Open Questions, docs/project/OPEN_QUESTIONS.md; unresolved items are interpreted together with later explicit decisions.